

Canada's Rough Fescue Grasslands

A Trial Restoration Project in Alberta is Yielding Encouraging Results

by Richard D. Revel

Volunteers transplant native sod to a suburban site—and their methods hold promise for other restoration work.

Rough fescue (*Festuca scabrella*) grassland covers approximately 1,400 square kilometers of southwestern Alberta, occurring on both the plains and the foothills of the southwestern part of the province (Strong and Leggat, 1981). Many different plant communities are present in this zone (Moss and Campbell, 1947; Looman, 1969, 1981). Because the rough fescue grassland zone abuts the moister aspen parkland zone to the northeast, trembling aspen (*Populus tremuloides*) communities commonly occur in depressions and on more northern aspects. On the drier southern aspects, species more common to the drier mixed-grass prairie found east of the rough fescue zone gradually increase in abundance. Soils are typically dark brown chernozems.

Looman (1969) notes that the productivity of native fescue grasslands is high when compared to other native grasslands, but still much lower than can be achieved with exotic grasses on the same land. He estimates that annual average productivity of fescue grassland is about 800 kg/ha, a figure that can be doubled or even tripled by conversion to brome grass (*Bromus inermis*), timothy (*Phleum pratense*), red fescue (*Festuca rubra*), and alfalfa (*Medicago sativa*). Because of their high potential for range and farming, most of the native fescue grasslands have been converted to agricultural use, and a great deal of research has been conducted on the effects of disturbance such as fire, haying, and grazing on these ecosystems (Looman, 1969; Bailey and

Anderson, 1978; Moss and Campbell, 1947). Twenty-four years ago Looman (1969) estimated that more than 90 percent of these grasslands had been greatly or moderately altered and suggested that "unless some suitable areas are placed in 'Nature Preserves' the time is not far off when the fescue grassland will have followed the true prairie into extinction." Unfortunately, alteration and elimination of these grasslands has not slowed, but continued relentlessly.

The continuing threat to these grasslands from both urban and exurban pressures lends a sense of urgency to fescue restoration research and has led to some interest and efforts to develop practical, effective ways of restoring rough fescue grassland communities. The Edgemont Project is one such effort.

The Restoration Site

Calgary, a city of some 700,000 people, is divided into many communities all under one municipal government. Each community has its own community association that, to a large degree, has considerable input into development and management issues within its boundaries via various association committees. These committees influence community development through direct lobbying, requests for advice from various city departments, and through elected alderpersons who represent all or part of several communities as defined by municipal electoral boundaries.



Raking the donor fescue grassland helps reduce thatch and allows for thicker cuts of sod with greater root mass.

The roots of the rough fescue project lie in what is known as the Native Species Parkland Project (NSPP) initiated in 1989 by an Edgemont Community Association committee dealing with community green space (Revel, 1991). Its goals as described by the NSPP committee (Edgemont Community Association, 1989) were to enhance the environment by helping nature establish a unique suburban native ecosystem. This in turn would promote environmental awareness, offer educational opportunities, enrich residents' lives, and provide an opportunity for people to get involved in "implementing and caring for their own unique landscape." The committee hoped that the project would serve as a prototype for other urban and suburban parks, although it is important to note that NSPP did not

specifically focus on grasslands but instead was committed to using native species of trees, shrubs, herbs, or grasses throughout the community rather than the more conventional exotic or agronomic species.

Following an initial productive spurt of enthusiasm, the project languished briefly because community leaders became involved in other pressing matters and also because no one involved in the project had a clear idea of how to achieve some of its goals. The NSPP deals with green space throughout the community but at the time I became involved the committee was most concerned with an area around the community center that was just being constructed. I agreed to review various ways of re-establishing the native grasslands around this site.

In doing this I evaluated four methods:

1. Seeding with a mix of commercially available non-natives similar enough to native species to appear native to the untrained eye.
2. Seeding with a mix of commercially available native species.
3. Seeding with native seed harvested by volunteers from native grasslands still existing near the restoration site.
4. Bringing in sod cut from native grassland of the appropriate type on a site scheduled for development.

I presented these alternatives for evaluation by our Native Species Parkland Project Committee. As a committee we rejected the first option because we were firmly committed to using only native species. We rejected the second option because only a small percentage of the native species that make up the rough fescue grasslands are commercially available and because seed that is available is prohibitively expensive and would not be representative of the local gene pool. We rejected option three, though acceptable from a biological perspective, because it would take a great deal of time and would require much specialized equipment for gathering and preparing the seed. We also felt that results would not be immediate enough to maintain volunteer interest, and we were concerned that growth would not be quick enough to suppress weed growth on the site or meet requirements of preventing site erosion in the short term. In addition, little is known about the appropriate pre-germination treatments for most of the vast array of species. We gave the last option preliminary acceptance on the grounds that if it was successful it would meet all requirements as set out in the project goals. The task of developing a technique for reconstructing the rough fescue grassland was left up to the author.

Criteria and Techniques

In developing a transplant technique consistent with the community's goals I established a number of design criteria specific to this project. The technique must:

1. Have a reasonable likelihood of success;
2. Be inexpensive;

3. Be suitable for undertaking by community volunteers;
4. Show immediate results and provide a sense of achievement and ownership by volunteers and the community at large;
5. Not require highly specialized equipment;
6. Make it possible to complete planting of several acres over one or two weekends, assuming reasonable volunteer participation (30 to 40 people);
7. Be relatively independent of weather conditions and the phenological stage of vegetation;
8. Be effective on steep slopes; and
9. Accommodate scientific analysis and long-term monitoring.

The first eight of these requirements were set down for reasons of expediency and were consistent with the practical goal of establishing a native rough fescue grassland on the project site. The last requirement was established in order to ensure that the project would provide adequate opportunity for careful scientific investigations pertaining to restoration, community change, and vegetation management over an extended period of time.

In May 1990, after locating several hundred acres of native sod slated for elimination by development, I examined three methods of stripping the sod and evaluated them against the design criteria established earlier. These methods included large twin-motor earth scrapers, Bobcats, and conventional light-duty sod cutters available from local rental agencies.

In trials with the large scrapers, I found they were able to cut sod up to 35 cm thick and would incorporate sufficient root and soil material from the fescue community to more or less ensure successful transfer of most species. But these scrapers did not cut an even sod thickness and could not empty intact sod from the scraper belly—the sod invariably emerged in an array of broken pieces. I rejected scrapers for these reasons as well as their cost, their inability to work on the steep slopes, and because they were inconsistent with a community volunteer-scale restoration project.

Initially, I undertook no trials with Bobcats as the small size of the bucket and

the cost of operator and equipment made this method unsuitable for restoring large areas quickly and inexpensively. My view was that they would be useful for excavation and planting of small trees and shrubs that are common in draws and depressions throughout the rough fescue grassland zone.

Fortunately, light-duty sod cutters worked better. When I tried one I found that an even sod up to a maximum of 5 cm thick was easily and quickly cut into manageable sizes for handling by volunteers. From a technical perspective cut and rolled sod was consistent with all design requirements. What was not clear, however, was whether such thin sod could be successfully transplanted and whether it included enough of the root system to enable plants to survive and overwinter. I had no information on root mass requirements to allow me to predict success or which species might survive the transplant. No evidence was found in the literature to provide guidance on these matters.

The Project Area

The Edgemont community is built on the western portion of Nose Hill, a geological

outlier established during the Rocky Mountain orogeny. Surficial geological material is predominantly a compact till although this has been overlain in some areas by lacustrine and outwash materials.

Elevational ranges within the community are in the order of 100 meters. The community has a network of ravines, some of which are protected as urban parkland largely because they are too steep for development, and others that have become locations for roads. The steep parkland ravines have substantial areas of native vegetation with fescue grasslands dominating the southern exposures and willow and aspen communities dominating the more northern aspects. Disturbed, but undeveloped, areas within and at the top of the ravine slopes have been reseeded to a mix of agronomic grasses conventionally used in the city. Brome grass, crested wheatgrass (*Agropyron cristatum*), and timothy are characteristic in the mix.

An inventory of the community revealed an area several hundred acres in size of undisturbed rough fescue grasslands that were slated for housing development within a three- to four-year period, although timing would depend upon housing demand. I selected one large patch of



Volunteers cut and roll native sod using light-duty cutters.

climax grassland less than 0.5 km from the proposed restoration as a donor site for restoration sod. This site, since developed into a housing estate, was located on flat to steeply sloping terrain, and the restoration site is on a steeply sloping ravine edge that had been disturbed and regraded during road construction and community facility development. The area selected for initial restoration efforts is approximately 0.6 ha (1.4 acres).

The rough fescue zone represents a major vegetation zone that is by no means uniform in its vegetation composition and incorporates many different plant communities. It is vital that the appropriate plant community for transfer be selected on a "site match" basis with the site to be restored. Failure to consider these ecological cross-matches would be a serious folly both from a conservation perspective as well as from economic and application perspectives. At this point we determined that the ecological match between our donor and restoration sites was good.

Initial Restoration Trials

In May 1991, as an initial trial to pre-test the viability of sod transplants, I cut and laid 20 m² of sod in a private yard over 0.75 to 1.5 m of topsoil. In May the spring growth is quite advanced, and I had concerns about how this would affect transplant success. I monitored the site for one year to determine whether the sod would survive and the species composition would be a good reflection of the original site.

My first observation was that the plants, which were growing rapidly at the time of transplanting, experienced a short check in growth and some die-back, and then resumed growth throughout the summer. At this stage all the original species of grasses, herbs, and shrubs seemed to flourish and in fact overall growth greatly exceeded that on the undisturbed donor site, probably because I removed accumulated thatch when I cut the sod and because the soil on the trial restoration site was richer, deeper, and had a greater capacity than that on the donor site. Altogether, approximately 35 native species survived the transplant experiments during the first summer, although the cover



Volunteers finish preparing the restoration site, while others lay sod.

values and dominance of these species at the transplant site varied considerably from those on the control site.

By April of the second year, although spring growth at that time was meager, there was sufficient evidence that the sod had survived to justify proceeding with the 0.6-ha community-based restoration project.

Planning the Community Project

Once our Native Species Parkland Committee, comprising five to ten involved residents, decided to proceed with the community-based restoration project, we had a great number of matters to attend to; pick the dates, publicize the event, organize equipment, and arrange food and celebratory events. In addition, I had to continue monitoring the trial plots to determine potential problems, and fine-tune technical matters related to sod cutting, transporting, and laying.

This phase of the project was one of high stress and a great deal of apprehension. We chose May 9 to May 11 as the dates for the community project, which gave us only one month to prepare. This decision fully committed us with no possibility of turning back for fear of losing community enthusiasm.

In order to whip up community spirit we described the project and provided contact names and phone numbers so volunteers could register or get further information in our local monthly community magazine. We also placed advertising boards at the major road entrances to Edgemont and undertook a phone campaign throughout the community to actively solicit volunteers. Once word of the event spread, various politicians, including our member of the Alberta Legislative Assembly, and city departments asked to participate.

The volunteer response was enormous. Initially we had a list of approximately 70 volunteers, which continued to grow as the day of the event drew near. Municipal and provincial politicians contacted us for information, endorsed the project, and offered to provide equipment, give speeches, and assist in other ways. Many private firms offered equipment, food, and drink free or at reduced rates, while volunteers offered pick-up trucks to haul the sod and promised to bring shovels, rakes, hoses, and other useful pieces of gardening equipment. Of great help were offers from one firm to donate two Bobcats and operators for the weekend and from two equipment-rental firms to provide sod cutters at reduced rates. We had not expected such generosity and enthusiasm

but it became obvious that we were going to be short neither equipment nor labor. By the time the project day arrived, we had a commitment of over 100 volunteers, four sod cutters, four pick-up trucks, a marquee, cooking facilities and a public address system.

Three concerns remained: the weather, site preparation, and the phenological stage of the native prairie. Nothing could be done about the weather except hope for a cool, overcast day and be prepared to cancel the project in the event of heavy rain.

Site preparation was to be handled by the City of Calgary, which had agreed to till and top-dress the restoration site with four inches of topsoil and treat it with a herbicide (Round-up) to diminish the chance of weed problems. The city also offered to try to install temporary irrigation from a local fire hydrant beside the site, although they forewarned us that the latter might not be possible.

The phenological stage of development at the time of transplanting became an increasing concern to me as the appointed day approached. The spring of 1991 was greatly delayed in comparison with the previous spring when the trial scale planting was done, and I had no empirical evidence that cutting and laying sod before green-up would produce satisfactory results (although one often moves plants during dormancy). My major concern related to the fact that cutting the sod would leave the majority of the roots and consequently stored food reserves behind at a time when such reserves are at their lowest point and the plants, just starting growth, are most in need of nutrients.

History and events overtook us, however, and we found ourselves so committed that we could not turn back. We simply went ahead and hoped for the best!

Project Implementation

On the evening of Friday, May 9, 1991, volunteers brought sod cutters to the donor site and cut several hundred square feet of sod. They left it in place there so it would be a simple matter to roll and transport it for an early start the next morning when additional volunteers ar-

rived. We experienced cutting difficulty because the cutter drive wheel slipped on the thick thatch from previous years of growth and non-use, but we resolved the problem by raking the dead grass with garden rakes before cutting. The raking also allowed us to cut deeper into the soil and gain a greater sod thickness and root mass.

On May 10th nearly 200 volunteers, ranging from the very young to the very old, arrived at the reconstruction site. The day was exceptionally hot and sunny. One group of volunteers set up the marquee, barbecue, and public address system while the remaining volunteers began the actual restoration work.

One volunteer group cut, rolled, and transported the sod to the restoration site, while another crew unloaded the trucks and began re-laying it in an interlocking pattern to reduce the chance of slippage in the event of heavy rain. Other volunteers selectively staked it in place as an extra precaution against slippage. Less than 45 minutes lapsed from the sod cutting to sod re-laying, so we had no concerns that it would be damaged by drying.

With an excess of volunteer labor and equipment I decided to interplant native shrubs and forbs that from my experience were less common in the native donor site but were common in the surrounding native fescue prairie. To achieve this I staked and flagged patches of the desired vegetation in the area surrounding the donor site, after which another crew set about digging them up with shovels, transporting them to the site and re-planting them in locations I staked at the restoration site.

By the time the sodding and the shrub/forb digging crews were fully organized and running smoothly under team leaders, the two Bobcats had arrived and needed an assignment. Part of the native fescue site had groves of trembling aspen trees and a variety of large willows growing in a site ecologically similar to the restoration site and I felt these would be attractive if interplanted with the sod at the latter. I assigned one Bobcat operator the task of digging up, transporting, and re-planting clumps of these species with as much soil and vegetation as possible. As the restoration site was steep-sloped and the only access was from the bottom, a

volunteer from the sod-laying group asked the second Bobcat operator to transport sod from the trucks to the top of the slope. The use of the Bobcat for this purpose made the whole sodding process proceed much more quickly with much less effort and a greatly reduced chance of back injury to volunteers.

The transplanting continued from 8 a.m. until 3 p.m., at which time ceremonies began with speeches from the local member of the Legislative Assembly of Alberta, a city alderperson, various community leaders, and a representative of the city Parks and Recreation Department. The Parks Department also planted and dedicated a number of white spruce trees on the site, although I considered this species a poor ecological match to the restoration site. By the end of the first day we had sodded approximately 0.4 ha (one acre).

Because the day was exceptionally hot, I was greatly concerned for sod survival as technical problems had prevented the city from installing irrigation. Fortunately, a local construction contractor working nearby took an interest in our project and volunteered to set up temporary irrigation using a local fire hydrant. This allowed volunteers to give the site a light watering with garden hoses after the sod was laid.

The project was scheduled for completion the next day. Overnight, however, a storm front moved into the area, bringing intense rains that forced us to cancel the day's efforts and reschedule completion. A weekend was selected in mid-June for continuation.

In mid-June approximately 25 volunteers and equipment were gathered, and between 9 a.m. and 2:30 p.m. we cut, transported, and laid approximately 0.2 ha (0.4 acres) of sod, completing the project. The day was much cooler than the first and with a smaller and more experienced crew of volunteers the work proceeded much more rapidly.

Follow-up and Ongoing Research

As indicated earlier, one of my goals for this project was to begin to build a base of experience and information useful in



By mid-summer of the second year, the transplanted fescue exhibits prolific seed production.

carrying out future rough fescue restoration efforts. To achieve this I have continued to monitor the sod to determine:

1. What species survived the transplant;
2. Weed infestation problems; and
3. General changes in species composition and abundance.

Concurrent with the above initial monitoring I have established permanent sample quadrants and will use them to compare:

1. Species composition and phenological differences and their causes.
2. Productivity of native and restored sites.
3. Competition among native and non-native species.
4. Reproductive success of native species.
5. Effects of different treatments such as mowing, burning, and trampling on the restored community.

Initial Findings

The following findings, though qualitative, are the result of careful observation of the three-year-old native rough fescue sod from the initial trial and the two-year-old sod from the larger community resto-

ration project through three and two growing seasons, respectively.

1. Plants in transplanted sod experienced an initial check in growth followed by continued growth after the sod rooted into the receiving site. More rapid growth occurs when the sod is laid over a well-topsoiled site such as at the initial trial site.
2. Species composition and abundance closely resembles that of the donor source climax community at the end of the first growing season. With time, the stability of initial species composition is altered and the plant community regresses to an earlier successional stage. This appears to be due to removal of root and rhizome masses and other below-ground parts deeper than five cm (19.5 in.) during sod cutting. Evidence for this is that transplanting clearly favored shallow-rooted species over deep-rooted species. For example, the shallower-rooted blugrass (*Poa* spp.) and wheatgrass (*Agropyron* spp.) have increased in abundance at the expense of rough fescue. In general, herbs are favored over grasses, probably in part for the same reason as suggested for the grasses.
3. Species diversity in the restored sites

was greater than in the undisturbed grasslands. I initially documented roughly 45 species at the donor site and have found 73 species at the restored site (see Table 1). Of these, about 20 are native and introduced weedy species, but some earlier successional species have appeared as well. These are probably from seed or vegetative parts in the sod or seed from the surrounding area for which the new site conditions and the transplantation process itself provided favorable conditions for establishment.

So far weed species have invaded the native sod mostly at the joints between sod. During the first year many annual weeds got established at these points such as burning bush (*Kochia scoparia*), tansey mustard (*Descurainia sophia*), lamb's quarters (*Chenopodium album*), and Russian thistle (*Salsola kali*), as did some of the perennial weedy grasses such as couch grass (*Agropyron repens*) and foxtail barley (*Hordeum jubatum*). The annuals have become less abundant in subsequent years as gaps between the sod have filled with leaf litter and the native species have infilled by lateral growth. Aggressive rhizomatous perennials that got established early on have been more persistent, however, and some have made incursions into the sod. Rhizomes of the couch grass remaining on the site before sodding got a stronghold in the cracks and during the first growing season began to penetrate the sod. It is too early to tell whether couch grass will be forced out by competition from the native species.

5. I initiated mowing trials on one patch of the restored sod in the fall and another patch in the spring. So far my observations indicate that fall mowing had little effect on species composition and abundance while mowing in the spring strongly favored wheatgrass and bluegrass species and disfavored rough fescue. Some herbs, such as goldenrod (*Solidago* spp.), bluebell (*Campanula* spp.), and vetch (*Lathyrus* spp.), also benefitted from mowing. As these species have colorful flowers many residents were pleased with their increased abundance.

Table 1: Species Comparisons Among Sites

Scientific Name	Common Name	Immediate Donor Site	Surrounding Donor Site	Restoration Site
<i>Achillea millifolium</i>	yarrow	y	y	y
<i>Agropyron cristatum</i>	crested wheatgrass	n	y	y
<i>Agropyron repens</i>	couchgrass	n	n	y
<i>Agropyron smithii</i>	western wheatgrass	y	y	y
<i>Agropyron subsecundum</i>	awned wheatgrass	y	y	y
<i>Agrostis scabra</i>	tickleglass	n	y	y
<i>Androsace septentrionalis</i>	ray pimpernel	n	y	y
<i>Anemone multifida</i>	prairie anemone	y	y	y
<i>Anemone patens</i>	prairie crocus	y	y	y
<i>Antennaria campestris</i>	pussy toes	y	y	y
<i>Arabis holboellii</i>	reflexed rock-cress	n	y	y
<i>Artemisia frigida</i>	fringed sage	y	y	y
<i>Artemisia ludoviciana</i>	pasture sage	n	y	y
<i>Aster laevis</i>	smooth aster	y	y	y
<i>Aster pansus</i>	prairie aster	y	y	y
<i>Astragalus agrestis</i>	milk vetch	y	y	y
<i>Bouteloua gracilis</i>	blue grama	y	y	y
<i>Bromus inermis</i>	brome	n	y	y
<i>Campanula rotundifolia</i>	hair bluebell	y	y	y
<i>Carduus nutans</i>	nodding thistle	n	n	y
<i>Carex filifolia</i>	thread-leaved sedge	y	y	y
<i>Castilleja lutea</i>	yellow paintbrush	y	y	y
<i>Cerastium arvense</i>	field chickweed	y	y	y
<i>Chenopodium album</i>	goosefoot	n	n	y
<i>Cirsium arvense</i>	Canada thistle	n	n	y
<i>Comandra umbellata</i>	bastard toad-flax	y	y	y
<i>Danthonia parryi</i>	parry oatgrass	y	y	y
<i>Descurainia sophia</i>	flixweed	n	n	y
<i>Dodecatheon conjugans</i>	shooting star	y	y	y
<i>Elymus canadensis</i>	Canada wildrye	y	y	y
<i>Erigeron caespitosus</i>	fleabane	y	y	y
<i>Festuca scabrella</i>	rough fescue	y	y	y
<i>Fragaria virginiana</i>	strawberry	y	y	y
<i>Gaillardia aristata</i>	brown-eyed susan	y	y	y
<i>Galium boreale</i>	northern bedstraw	y	y	y
<i>Geranium richardsonii</i>	white geranium	n	y	y
<i>Geranium viscosissimum</i>	sticky purple geranium	y	y	y
<i>Geum triflorum</i>	three-flowered avens	y	y	y
<i>Heuchera richardsonii</i>	alum root	y	y	y
<i>Hordeum jubatum</i>	foxtail barley	n	y	y
<i>Kochia scoparia</i>	burning bush	n	n	y
<i>Koeleria cristata</i>	june grass	y	y	y
<i>Lappula echinata</i>	stickweed	n	n	y
<i>Lathyrus ochroleucus</i>	cream-coloured vetch	y	y	y
<i>Lilium philadelphicum</i>	western prairie lily	y	y	y
<i>Lithospermum ruderales</i>	puccoon	y	y	y
<i>Melilotus alba</i>	white sweetclover	n	n	y
<i>Melilotus officinalis</i>	yellow sweetclover	n	n	y
<i>Oxytropis sericea</i>	early yellow loco-weed	y	y	y
<i>Oxytropis splendens</i>	showy loco-weed	y	y	y
<i>Phleum pratense</i>	timothy	n	n	y

Table 1 (continued): Species Comparisons Among Sites

Scientific Name	Common Name	Immediate Donor Site	Surrounding Donor Site	Restoration Site
<i>Poa secunda</i>	bluegrass	y	y	y
<i>Potentilla fruticosa</i>	shrubby cinquefoil	y	y	y
<i>Potentilla gracilis</i>	green cinquefoil	n	y	y
<i>Raphanus raphanistrum</i>	jointed charlock	n	n	y
<i>Ribes oxycanthoides</i>	northern gooseberry	n	y	y
<i>Rosa woodsii</i>	Wood's rose	y	y	y
<i>Salsola kali</i>	Russian thistle	n	n	y
<i>Sisyrinchium montanum</i>	blue-eyed grass	y	y	y
<i>Smilacina stellata</i>	star-flowered Solomon's seal	y	y	y
<i>Solidago missouriensis</i>	low goldenrod	y	y	y
<i>Sonchus arvensis</i>	sow thistle	n	n	y
<i>Stipa comata</i>	needle and thread grass	y	y	y
<i>Symphoricarpos occidentalis</i>	buck brush	y	y	y
<i>Taraxacum officinale</i>	dandelion	n	y	y
<i>Thlaspi arvensis</i>	stinkweed	n	n	y
<i>Thalictrum occidentale</i>	western meadow rue	y	y	y
<i>Thermopsis rhombifolia</i>	golden bean	y	y	y
<i>Tragopogon dubius</i>	goats beard	n	y	y
<i>Trifolium hybridum</i>	clover	n	n	y
<i>Vicia americana</i>	american vetch	y	y	y
<i>Vicia sparsifolia</i>	narrow-leaved vetch	y	y	y
<i>Viola adunca</i>	early blue violet	y	y	y

6. Spring green-up occurred approximately two weeks earlier on the sodded site than on comparable sites in the undisturbed control site and other fescue grasslands on similar ecological sites in Edgemont. This can be attributed to a greatly diminished thatch cover due to raking on the reconstructed sites, which allows more rapid warming of the soils. In support of this conclusion, I found that the plot on the restored site that I burned in the early spring greened-up even earlier, while sites mowed in the spring and covered with a layer of clippings greened-up slowly.
7. So far I have found no difference in the performance of sods taken before spring green-up, just as green-up was occurring, or in mid-June when the vegetation was well-developed.

In conclusion, the sodding technique holds considerable promise for other small- to medium-scale restoration pro-

jects and provides a means for preserving native sod that otherwise would be destroyed. Our work so far also illustrates the educational, aesthetic, and conservation benefits of restoration projects carried out by volunteers in urban parks.

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